



Linear Models Homework and Exit Ticket

Grade 8 • Mathematics
Homework

Original Practice • Assessment • Complete Teacher Key



Teacher Guide

Level: Grade 8

State: North Carolina

Subject: Mathematics

Resource type: Homework

Estimated time: 40-50 minutes

Standard / objective: NC.8.F.4 - Analyze functions that model linear relationships; determine and interpret rate of change and initial value from two values or from a graph, table, or description.

Source note: North Carolina Department of Public Instruction, Mathematics Standards Vertical Progression:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>; NC Check-Ins 2.0 Mathematics

Specifications: <https://www.dpi.nc.gov/documents/accountability/testing/nc-checkin/nc-check-ins-20-mathematics-grades-3-8-specifications> (accessed September 14, 2026).

What is included: branded cover, teacher guide, student directions, 16 original practice items, 5-item exit ticket, complete answer key/worked guidance, differentiation supports, printing notes, and license statement.

Purpose: Build linear models from tables and contexts and interpret slope and initial value with units.

Preparation: Print student pages single-sided or assign through a secure LMS. Review the key and select tools allowed for your learners. No special commercial curriculum is required.

Access and differentiation

AIG: require a second representation, proof, counterexample, transfer task, or student-created extension. **504:** provide extended time, chunked sections, breaks, reduced copying, and accessible digital zoom as specified by the plan. **EC:** preteach vocabulary, model one item, provide manipulatives/visuals, and accept aligned response modes from the IEP. **ESL:** preview key terms, provide bilingual resources and sentence frames, and allow partner rehearsal. Use each learner's current plan; this guide does not replace it.



Student Practice

Learning focus: Build linear models from tables and contexts and interpret slope and initial value with units.

Directions: Complete all items. Show calculations, cite evidence, or explain your reasoning when requested. Check your work before submitting.

1. Find slope through (2,7) and (6,19).
2. Write $y=mx+b$ when $m=4$ and $b=-2$.
3. Table: x 0,2,4; y 5,11,17. Find m and b .
4. A taxi charges \$4 plus \$2.50 per mile. Model.
5. Interpret 4 in the taxi model.
6. Find taxi cost for 8 miles.
7. Line through (3,10) with slope 2. Find b .
8. Compare $y=3x+1$ and $y=3x-5$.



Student Practice - Continued

9. A tank has 90 L and drains 6 L/min. Model.

10. When is tank at 42 L?

11. Does (5,16) lie on $y=3x+1$?

12. From y values 8,13,18 as x rises by 1, find rate.

13. Write context for $y=12x+30$.

14. Explain negative slope in context.

15. Find equation through (0,-3),(4,5).

16. State one limitation of a linear model.



Exit Ticket

Complete independently in 8-10 minutes.

1. State the main skill or concept in your own words.
2. Create one accurate example that uses today's skill.
3. Explain how you checked your example.
4. Identify one common error and correct it.
5. Rate your confidence from 1-4 and name your next practice step.

Quick rubric: 4 = accurate, complete, and well justified; 3 = accurate with minor omission; 2 = partial understanding; 1 = limited evidence; 0 = blank or unrelated.



Complete Teacher Answer Key

1. 3
2. $y=4x-2$
3. $m=3$, $b=5$
4. $C=2.5m+4$
5. \$4 initial fee
6. \$24
7. 4
8. Same rate; first starts 6 units higher
9. $V=90-6t$
10. 8 minutes
11. Yes
12. 5
13. Answers vary; \$30 start plus \$12 per unit
14. Output decreases as input increases
15. $y=2x-3$
16. Constant rate may not continue outside observed range.

Exit ticket guidance: Responses vary. Award full credit when the student accurately explains the focus, supplies a correct example, describes a valid check, identifies and corrects a realistic misconception, and chooses a specific next step.

Verification: Fixed responses and calculations were checked during production. Open responses are evaluated against the criteria above.

Originality and use: This packet contains original educational content. It does not reproduce released-test items or branded curriculum. One purchaser may use it with their own students; redistribution, resale, and public posting are not permitted.